Demand and supply

Lecture notes

What this lecture is about. A market has two sides. Buyers decide how much to buy at each price; sellers decide how much to sell. This lecture builds both sides from simple pieces: the most each buyer would pay, and the least each seller would accept. We then put the two sides together. In a small market with ten buyers and ten sellers we ask who should trade, at what price, and who gets the gain from trade. Finally we find the equilibrium of a market from two equations, and we see what happens when a price is held below or above it.

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1 Demand for a single person

Start with one person and one good. You can subscribe to several streaming services, and each costs €10 a month. They are not all worth the same to you. Sort them from the one you value most to the one you value least:

subscription	1st	2nd	3rd	4th
worth to you (a month)	€16	€12	€8	€4

Question: how many subscriptions do you keep?

Each subscription costs €10 a month. The 1st is worth €16 a month to you, the 2nd €12, the 3rd €8 and the 4th €4. How many do you keep?

Step 1: the rule. Keep one more subscription as long as it is worth at least its price, €10.

Step 2: go down the list. The 1st is worth €16 $\geq$ €10: keep it. The 2nd is worth €12 $\geq$ €10: keep it. The 3rd is worth only €8 $<$ €10: drop it. The 4th is worth even less.

You keep **2** subscriptions.

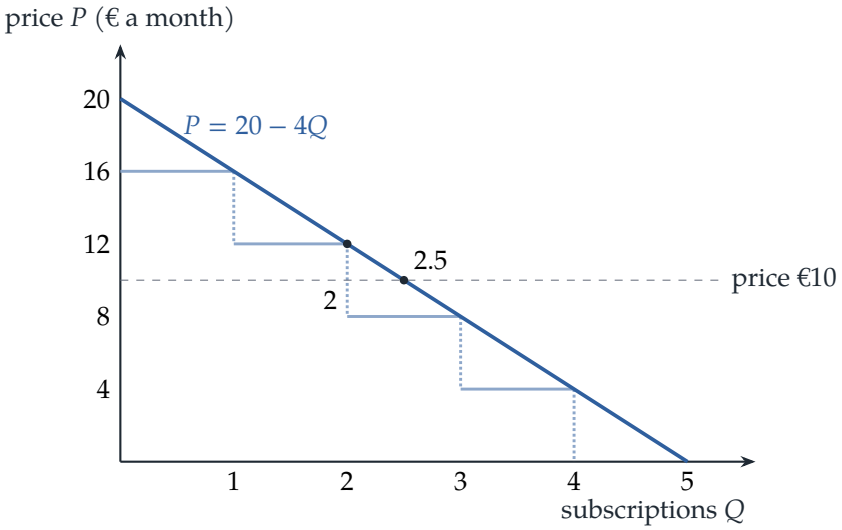
Two demand curves. The table gives a demand curve in the form of a staircase. Put the number of subscriptions Q on the horizontal axis and the price P on the vertical axis. The first step is at height €16, the second at €12, the third at €8 and the fourth at €4. At any price, the quantity you demand is the number of steps that are at or above that price.

The values fall by €4 each time, so a straight line fits the staircase exactly:

$$P = 20 - 4Q.$$

At $Q = 1, 2, 3, 4$ the line gives $16, 12, 8, 4$: it passes through the right-hand corner of each step. The number 20 is the price at which you would keep none, read on the line (on the staircase you already keep none at any price above €16).

At a price of €10 the staircase gives 2 subscriptions. The line gives $10 = 20 - 4Q$, so $Q = 2.5$. The straight line is a smooth summary of the steps; it rounds them.



The staircase (light blue) and the straight line $P = 20 - 4Q$ (dark blue). At €10: 2 on the staircase, 2.5 on the line.

2 Market demand from reservation prices

Reservation price

A buyer's **reservation price** is the most she would pay for the good (here: a month of a subscription).

Now consider many buyers, each of whom wants at most one unit, for example one month of an AI chatbot subscription. Each buyer has her own reservation price. To build the market demand curve:

1. **Sort the buyers** from the highest reservation price to the lowest, and give each one a rank: 1, 2, 3, ...
2. **Apply the rule.** At a given price, everyone whose reservation price is at least the price buys. The number of buyers is the quantity demanded at that price.
3. **Draw the bars.** Put one bar per buyer, in rank order, with height equal to her reservation price. The tops of the bars form a falling staircase: this staircase is the market demand curve.

The rank in line *is* the quantity. The 20th subscription goes to the 20th buyer in line, and the price at which the 20th subscription is sold is that buyer's reservation price. A straight line drawn through the tops of the bars summarises the staircase, as in section 1. In class, the bars

came from the reservation prices that students had given in lecture 1. The same method works for any list of buyers; section 6 applies it to ten buyers with values from €100 down to €10.

3 Demand and inverse demand

A demand curve can be read in two directions.

Demand

Demand: what quantity do consumers want to buy at a given price?

Read *horizontally*: pick a price on the vertical axis, go across to the curve, and read the quantity below. For the streaming subscriptions of section 1, at a price of €10 the demand is 2 subscriptions (on the staircase).

Inverse demand

Inverse demand: what price are consumers willing to pay for a given quantity?

Read *vertically*: pick a quantity on the horizontal axis, go up to the curve, and read the price. On the line $P = 20 - 4Q$, for the 2nd subscription the price is $20 - 4 \cdot 2 = €12$. Another way to read it: the buyer of the last unit, the **marginal buyer**, is willing to pay exactly that price. The inverse demand curve shows the **marginal willingness to pay** at each quantity.

Law of demand

Price ↑ ⇒ quantity demanded ↓ (all else equal).

- **Demand form.** When the price rises, and nothing else changes, the quantity demanded falls.
- **Inverse form.** The marginal willingness to pay decreases as the quantity increases. To sell one more unit, the price must fall to the next buyer in line, whose reservation price is lower (the marginal buyer).
- **The picture.** A downward-sloping curve.

The words "all else equal" matter. They hold fixed everything except the price of the good, such as income and the prices of other goods. When one of those changes, the whole curve shifts; that comes in a later lecture.

4 Demand functions

A demand function writes demand as an equation, $Q_D = f(P)$. Take an AI subscription, with the price P in € per month and the quantity Q in thousand subscribers:

$$Q = 120 - 4P.$$

- **The 120.** If it were free ($P = 0$), 120,000 people would subscribe.
- **The $-4P$.** Each €1 more a month means 4,000 fewer subscribers.
- **At €20.** $Q = 120 - 4 \cdot 20 = 40$: 40,000 subscribers.

4.1 From demand to inverse demand

The inverse demand function solves the demand function for P. The method is always the same: move the P-term to one side, then divide *every* term by its coefficient.

Question 4.1: the AI subscription

Demand is $Q = 120 - 4P$ (Q in thousands of subscribers). What is the inverse demand? Then draw it.

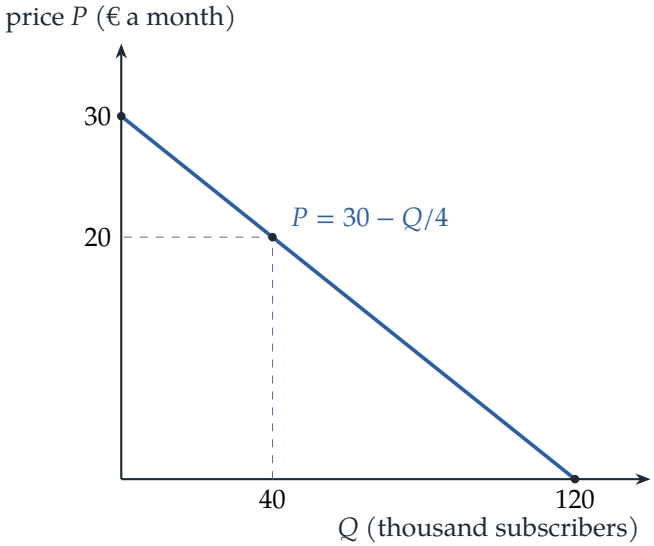
Step 1: move the P-term. $Q = 120 - 4P$ becomes $4P = 120 - Q$.

Step 2: divide every term by 4. $P = 30 - Q/4$.

Step 3: read it. The 30: at €30 or more, nobody subscribes. The $-Q/4$: to win 4,000 more subscribers, the price must fall by €1.

Step 4: draw it. The line meets the price axis at €30 ($Q = 0$) and the quantity axis at 120 ($P = 0$). Join the two points.

$P = 30 - Q/4$. Common mistakes: dividing only one term by 4 (which gives $P = 120 - Q/4$), or swapping the roles of the numbers (which gives $P = 30 - 4Q$), or a sign error.



Demand for the AI subscription. It meets the price axis at €30 and the quantity axis at 120. At €20, 40 (thousand) subscribe.

The deck then gives six more questions of the same kind.

Question 4.2: a music-streaming plan in Austria

P in € per month, Q in thousand subscribers. Demand is $Q = 100 - 5P$. What is the inverse demand? Then draw it.

$Q = 100 - 5P \Rightarrow 5P = 100 - Q \Rightarrow P = 20 - Q/5$, dividing *every* term by 5. Check: $Q = 0$ gives $P = 20$ in both forms. The 20 means that at €20 or more nobody buys. To draw it: price axis at €20, quantity axis at 100; join the two points.

$P = 20 - Q/5$.

Question 4.3: an AI tutor app

P in € per month, Q in thousand users. Demand is $Q = 60 - 3P$. Where does the demand curve meet the price axis and the quantity axis?

Price axis: there nobody buys, $Q = 0$, so $0 = 60 - 3P$, $3P = 60$ and $P = 20$. **Quantity axis:** there the app is free, $P = 0$, so $Q = 60$, that is 60,000 users. Join the two points with a straight line.

The price axis at €20, the quantity axis at 60 (60,000 users).

Question 4.4: printing at a print shop in Vienna

P in cents per page, Q in pages per day. Demand is $Q = 160 - 4P$. The 160: if printing were free, customers would print 160 pages a day. The $-4P$: each extra cent per page means 4 fewer pages a day. What is the inverse demand? Then draw it.

$Q = 160 - 4P \Rightarrow 4P = 160 - Q \Rightarrow P = 40 - Q/4$. Check: $Q = 0$ gives $P = 40$ in both forms. Read it: at 40 cents a page or more, nobody prints; to sell 4 more pages a day, the price must fall by 1 cent. To draw it: price axis at 40 cents, quantity axis at 160 pages; join the two points.

$P = 40 - Q/4$.

Question 4.5: monthly memberships at Vienna gyms

P in € per month, Q in members. Demand is $Q = 1600 - 40P$. What is the inverse demand? Then draw it.

$Q = 1600 - 40P \Rightarrow 40P = 1600 - Q \Rightarrow P = 40 - Q/40$. Check: $Q = 0$ gives $P = 40$; at €40 or more nobody buys. To draw it: price axis at €40, quantity axis at 1600; join the two points.

$P = 40 - Q/40$.

Question 4.6: day passes at Vienna gyms

P in € per pass, Q in passes per day. Demand is $Q = 240 - 6P$. Where does the demand curve meet the price axis and the quantity axis?

Price axis: $Q = 0$, so $6P = 240$ and $P = 40$. **Quantity axis:** $P = 0$, so $Q = 240$. Join the two points with a straight line.

The price axis at €40, the quantity axis at 240.

Question 4.7: an AI writing-assistant app

P in € per month, Q in thousand users. Demand is $Q = 150 - 2.5P$. What is the inverse demand? Then draw it.

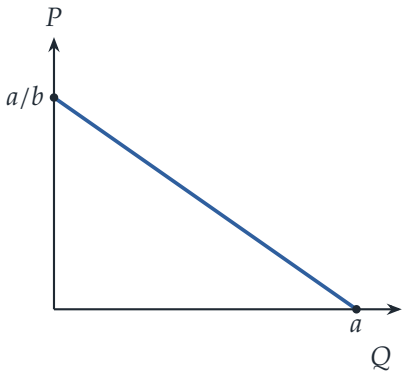
$Q = 150 - 2.5P \Rightarrow 2.5P = 150 - Q \Rightarrow P = 60 - 0.4Q$, dividing every term by 2.5 ($150/2.5 = 60$ and $1/2.5 = 0.4$). Check: $Q = 0$ gives $P = 60$; at €60 or more nobody buys. To draw it: price axis at €60, quantity axis at 150; join the two points.

$P = 60 - 0.4Q$.

4.2 Drawing any linear demand curve

Every linear demand function has the form

$$Q = a - bP.$$



It is drawn from its two intercepts:

- **Price axis:** $Q = 0$ gives $P = a/b$. This is the **choke price**: at this price or above, nobody buys.
- **Quantity axis:** $P = 0$ gives $Q = a$, the quantity if the good were free.

Join the two points with a straight line. For $Q = 120 - 4P$: $a/b = 120/4 = 30$ and $a = 120$, as in question 4.1.

5 Supply

Law of supply

Price ↑ ⇒ quantity supplied ↑.

- **Any firm.** Producing more needs overtime, extra staff, new machines: each extra unit costs more. This is called rising marginal cost.
- **Miners.** When the price is low, mining barely pays the electricity, so fewer miners mine.

5.1 Ten miners

Ten miners each have one machine. The machines are the same; what differs is the electricity bill. Sorted from the cheapest:

miner	1	2	3	4	5	6	7	8	9	10
cost of mining (€)	200	250	300	350	400	450	500	550	600	650

Question 5.1: how many miners mine?

The price is €460. How many miners switch on their machines and mine?

Step 1: the rule. A miner mines when the price covers her cost.

Step 2: go along the list. Miners 1 to 6 have costs from €200 up to €450, all at most €460: they mine. Miner 7 needs €500, which is more than €460: she does not.

6 miners mine.

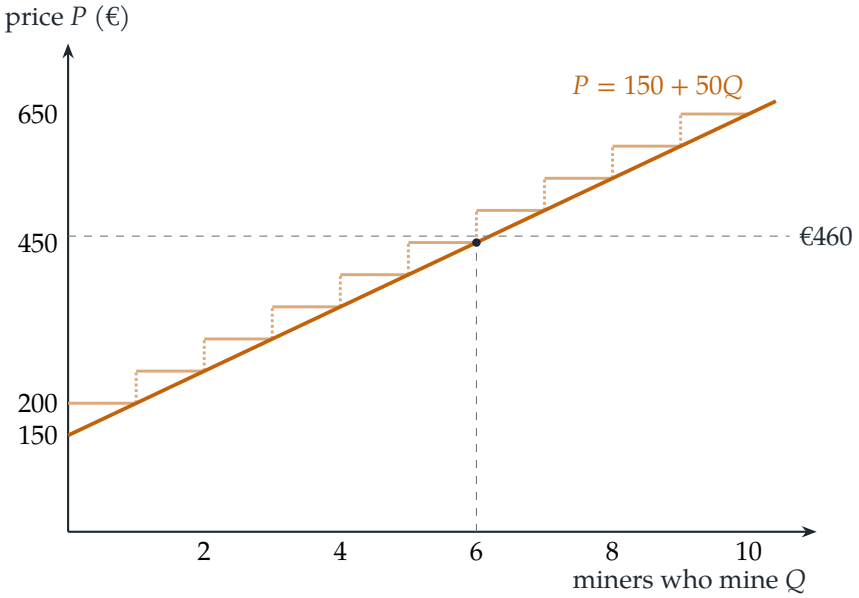
Supply

Supply: what quantity do sellers want to sell at a given price?

The costs rise by €50 from one miner to the next, and miner n has cost $150 + 50n$. A straight line through the steps is

$$Q = \frac{P}{50} - 3 \qquad (\text{for } P \geq €150),$$

where Q is the number of miners who mine. At €450: $Q = 450/50 - 3 = 9 - 3 = 6$. So at €450, 6 miners mine.



The ten miners as a staircase (light orange): step n is at the cost of miner n. The line $P = 150 + 50Q$ passes through the right-hand corner of each step. At €460, 6 miners mine.

5.2 From supply to inverse supply

Inverse supply

Inverse supply: what price do sellers need to offer a given quantity?

Question 5.2: the miners

Supply is $Q = P/50 - 3$. What is the inverse supply? Then draw it.

Step 1: bring -3 across. It becomes $+3$: $Q + 3 = P/50$.

Step 2: multiply every term by 50. $P = 50Q + 150$. Keep the sign.

Step 3: read it. The 150: at €150 or less, nobody mines (on the line). The 50: each further miner needs a €50 higher price.

$P = 150 + 50Q$. Common mistakes: losing the sign ($P = 50Q - 150$), dividing instead of multiplying ($P = 3 + Q/50$), or swapping the intercept and the slope ($P = 150Q + 50$).

Read vertically, as with demand: for 6 miners to mine, the price must be $150 + 50 \cdot 6 = €450$. Another way to read it: the 6th miner, the **marginal miner**, needs €450 to switch on her machine. The inverse supply curve shows the cost of the marginal seller at each quantity.

The deck then gives six more questions of the same kind.

Question 5.3: freelance video editors on a jobs platform

P in € per hour, Q in hundred hours per week. Supply is $Q = 4P - 100$. What is the inverse supply? Then draw it.

Bring -100 across (it becomes $+100$): $4P = Q + 100$. Divide every term by 4: $P = 25 + Q/4$. Check: $P = 25$ gives $Q = 0$, so below €25 nobody offers anything. To draw it: start at €25 on the price axis; one more point: at €30, $Q = 4 \cdot 30 - 100 = 20$.

$P = 25 + Q/4$.

Question 5.4: seats of an AI note-taking tool sold by resellers

P in € per seat per month, Q in thousand seats. Supply is $Q = 5P - 40$. Where does the supply curve start on the price axis, and how many seats are offered at €12?

Start: $Q = 0$ gives $5P = 40$, so $P = 8$. Below €8 nobody offers anything. **One more point:** at €12, $Q = 5 \cdot 12 - 40 = 60 - 40 = 20$, that is 20,000 seats. Join the two points.

It starts at €8; at €12, 20,000 seats are offered.

Question 5.5: dog-walking by students

P in € per walk, Q in walks per week. Supply is $Q = 5P - 15$. What is the inverse supply? Then draw it.

Bring 15 to the other side: $5P = Q + 15$. Divide every term by 5: $P = 3 + Q/5$. Check: at $P = 3$ nobody offers a walk ($Q = 0$). To draw it: start at €3 on the price axis; one more point: at €5, $Q = 25 - 15 = 10$.

$P = 3 + Q/5$.

Question 5.6: cloud storage plans

P in € per month, Q in thousand plans. Supply is $Q = 10P - 30$. What is the inverse supply? Then draw it.

Bring -30 across: $10P = Q + 30$. Divide every term by 10: $P = 3 + Q/10$. Check: $P = 3$ gives $Q = 0$, so below €3 nobody offers anything. To draw it: start at €3 on the price axis; one more point: at €5, $Q = 50 - 30 = 20$.

$P = 3 + Q/10$.

Question 5.7: satellite launch slots on reusable rockets

P in € million per slot, Q in slots per year. Supply is $Q = 2P - 30$. Where does the supply curve start on the price axis, and how many slots are offered at €35 million?

Start: $Q = 0$ gives $2P = 30$, so $P = 15$. Below €15 million nobody offers anything. **One more point:** at €35 million, $Q = 2 \cdot 35 - 30 = 70 - 30 = 40$. Join the two points.

It starts at €15 million; at €35 million, 40 slots are offered.

Question 5.8: sponsored posts by Instagram creators

P in € per post, Q in posts per week. Supply is $Q = 0.5P - 60$. What is the inverse supply? Then draw it.

Bring -60 across: $0.5P = Q + 60$. Divide every term by 0.5, which is the same as multiplying by 2: $P = 120 + 2Q$. Check: $P = 120$ gives $Q = 0$, so below €120 nobody offers anything. To draw it: start at €120 on the price axis; one more point: at €200, $Q = 100 - 60 = 40$.

$P = 120 + 2Q$.

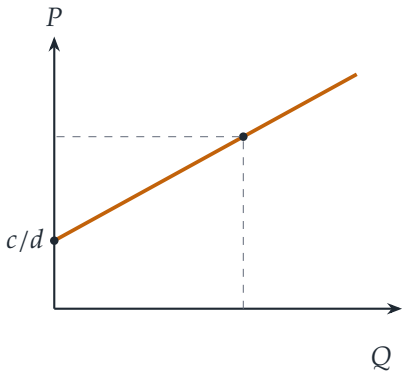
5.3 Drawing any linear supply curve

A linear supply function has the form

$$Q = dP - c.$$

- **Price axis:** $Q = 0$ gives $P = c/d$. Below this price nobody sells.
- **One more point:** pick a higher price and compute Q.

Join the two points. A supply line of this kind has no crossing with the quantity axis in the picture: it starts on the price axis and rises. For $Q = 4P - 100$: it starts at $100/4 = €25$, and at €50, $Q = 200 - 100 = 100$.



6 A synthetic market

We now put buyers and sellers together in a small market built for the purpose. There are 10 buyers and 10 sellers. Each buyer wants one unit; each seller has one unit.

	1	2	3	4	5	6	7	8	9	10
buyers: the most they would pay	€100	€90	€80	€70	€60	€50	€40	€30	€20	€10
sellers: the least they would accept	€10	€20	€30	€40	€50	€60	€70	€80	€90	€100

The question. Who should trade with whom, and at what price?

Gain of a trade

The gain of a trade is the buyer's value minus the seller's cost.

We try different matching rules (who trades with whom) and then different price rules (at what price).

6.1 Matching rule 1: same with same

Rule: each buyer trades with the seller whose cost equals her value, at that price. The €100 buyer trades with the €100 seller, the €90 buyer with the €90 seller, and so on down to €10 with €10. Every gain is 0.

Question 6.1: same with same

How much value do these 10 trades create?

Each buyer pays exactly what the good is worth to her, and each seller gets exactly his cost. So each gain is 0, and the total is 0.

€0. Ten trades, and nobody is better off: more trades is not the same as more value.

6.2 Matching rule 2: best with best

Rule: the buyer who values the good most trades with the cheapest seller, the second with the second, and so on.

buyer	€100	€90	€80	€70	€60	€50	€40	€30	€20	€10
seller	€10	€20	€30	€40	€50	€60	€70	€80	€90	€100
gain	90	70	50	30	10	−10	−30	−50	−70	−90

Question 6.2: best with best

How many of these pairs should trade?

Step 1: look at the gains. The first 5 pairs have positive gains. From the 6th pair on, every gain is negative ($-10, -30, -50, -70, -90$): those trades would destroy value.

Step 2: add up the gains of the first 5 pairs. $90 + 70 + 50 + 30 + 10 = 250$.

Step 3: can another matching do better? Any 5 trades use 5 buyers and 5 sellers. The total gain is the buyers' values minus the sellers' costs. It is largest with the 5 highest values ($100 + 90 + 80 + 70 + 60 = 400$) and the 5 lowest costs ($10 + 20 + 30 + 40 + 50 = 150$): $400 - 150 = 250$. A 6th trade would add a buyer worth at most €50 and a seller costing at least €60, so it would lower the total.

5 pairs, with a total gain of **€250**. No other matching creates more than €250.

6.3 Price rule 1: all gain to the sellers

The 5 best pairs trade. Rule: every pair trades at the buyer's value.

pair (value–cost)	100–10	90–20	80–30	70–40	60–50
price	€100	€90	€80	€70	€60
seller gains	90	70	50	30	10
buyer gains	0	0	0	0	0

Question 6.3: price at the buyer's value

What is the total value created by these trades?

Seller gain = price − cost; buyer gain = value − price = 0. Sellers gain $90+70+50+30+10 = 250$; buyers gain 0.

€250, the same as with any other prices. All of it goes to the sellers (the average price is $(100 + 90 + 80 + 70 + 60)/5 = €80$).

6.4 Price rule 2: all gain to the buyers

Rule: every pair trades at the seller's cost.

pair (value–cost)	100–10	90–20	80–30	70–40	60–50
price	€10	€20	€30	€40	€50
seller gains	0	0	0	0	0
buyer gains	90	70	50	30	10

Question 6.4: price at the seller's cost

What is the total value created by these trades?

Buyers gain $90 + 70 + 50 + 30 + 10 = 250$; sellers gain 0.

€250 again. Now all of it goes to the buyers (the average price is $(10+20+30+40+50)/5 =$ €30). The price only decides who gets the €250.

6.5 Price rule 3: any price in between

Rule: each pair agrees on some price between the seller's cost and the buyer's value.

Question 6.5: one pair, two prices

The buyer with value €100 and the seller with cost €10 agree on a price of €40. How much do the two gain together? Now they agree on €60 instead. How much do they gain together?

At €40. Buyer: $100 - 40 = 60$. Seller: $40 - 10 = 30$. Together: $60 + 30 = 90 = 100 - 10$.

At €60. Buyer: $100 - 60 = 40$. Seller: $60 - 10 = 50$. Together: $40 + 50 = 90$.

€90 at both prices. Any price between €10 and €100 gives the same €90; only the split changes.

In symbols: with value v, cost c and price p, the buyer gains $v - p$ and the seller gains $p - c$. Together they gain $(v - p) + (p - c) = v - c$. The price cancels.

Insight: the price splits the gain

The total gain is the same at any price. With price rule 1 (the buyer's value), the sellers get all €250 and the buyers get nothing. With price rule 2 (the seller's cost), the buyers get all €250 and the sellers get nothing. With any price in between, the €250 is shared. So the price does not change how much value trade creates; it decides how much goes to the sellers and how much to the buyers.

6.6 One price for everyone

In a market, everyone pays the same price. At each price:

- buyers with value $\geq$ price want to buy;
- sellers with cost $\leq$ price want to sell;
- the most eager buyers and the cheapest sellers trade, and the number of trades is the smaller of the two numbers.

At exactly €50 the €50 buyer, and at exactly €60 the €60 seller, does not care whether to trade (gain 0). The table counts them in (6). If they stay out, 5 buy and 5 sell.

price	€30	€40	€50	€55	€58	€60	€70	€80
buyers who want to buy (value ≥ price)	8	7	6	5	5	5	4	3
sellers who want to sell (cost ≤ price)	3	4	5	5	5	6	7	8
trades (the smaller number)	3	4	5	5	5	5	4	3
total value created (€)	210	240	250	250	250	250	240	210

How the last row is computed: at €30, 3 trades take place, between the 3 highest-value buyers and the 3 cheapest sellers: $(100 + 90 + 80) - (10 + 20 + 30) = 270 - 60 = 210$. At €40, 4 trades: $(100 + 90 + 80 + 70) - (10 + 20 + 30 + 40) = 340 - 100 = 240$. The columns for €70 and €80 are the mirror image. Away from the middle, one side of the market is short, and value is lost.

Question 6.6: at which prices do buyers equal sellers?

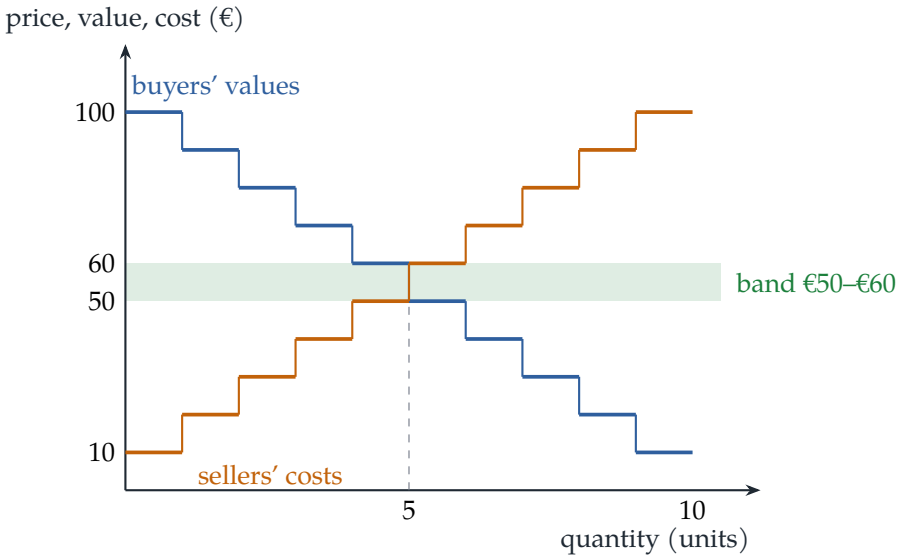
Draw the buyers in blue, from the highest value down, and the sellers in orange, from the lowest cost up. At which prices is the number of buyers equal to the number of sellers? Options: only at €55; any price from €50 to €60; any price from €30 to €80; at no price.

Step 1: inside the band. At any price strictly between €50 and €60, the 5 buyers with values €100 to €60 want to buy, and the 5 sellers with costs €10 to €50 want to sell. Exactly the 5 best pairs trade, and €250 is created.

Step 2: the ends of the band. At exactly €50 the €50 buyer is indifferent; at exactly €60 the €60 seller is indifferent. If they stay out, buyers and sellers are still 5 and 5.

Step 3: outside the band. Below €50 there are more buyers than sellers; above €60 there are more sellers than buyers.

Any price from €50 to €60. The market clears at every price in this band, not at one single price (€55 is only the middle of the band).



The synthetic market. Demand (blue) is the buyers' values from the highest down; supply (orange) is the sellers' costs from the lowest up. They meet at 5 units, anywhere in the shaded band from €50 to €60.

7 A trading floor

In class, the same synthetic market was run as a trading floor with 20 AI traders: 10 AI buyers (values €10 to €100) and 10 AI sellers (costs €10 to €100). Each trader is a computer program driven by a language model. The rules of the floor were:

1. Each buyer wants **one** unit and has a private **value**: the most she would pay. Each seller has **one** unit and a private **cost**: the least he would accept.
2. Nobody knows anyone else's number. The traders only see one shared chat.
3. In each round a few traders, chosen at random, may speak. A buyer can post an offer to buy ("buy at €..."), called a **bid**); a seller can post an offer to sell ("sell at €..."), called an **ask**); anyone can **accept** the best offer on the floor, or **pass**.
4. A new bid must be higher than the best bid so far; a new ask must be lower than the best ask so far.
5. A **trade** happens when a bid meets an ask, or when someone accepts the best offer. It takes place at the price that was already on the floor. Then the floor is cleared, and offers start fresh.
6. Nobody may trade at a loss (a referee program blocks it). Each trader trades at most one unit per period.
7. Gains: buyer = value − price; seller = price − cost. No trade means no gain.
8. There are 4 periods with the same numbers each time, so the traders can learn.

What the theory predicts. The traders do not know the curves. Section 6 still tells us where the market should end up: 5 trades, between the 5 buyers with values €100 to €60 and the 5 sellers with costs €10 to €50, at prices in the band from €50 to €60. The total gain is then the maximum, €250, whoever pairs with whom and at whatever price. A trade with a buyer or a seller from outside these two groups can still give both of them a gain, but it takes the place of someone inside the group, so the total gain of the period falls below €250. The experiment asks whether prices move towards the band even though nobody knows the curves.

8 Equilibrium

Equilibrium

Equilibrium: demand and supply give the same quantity and the same price.

In a picture, the equilibrium is where the demand curve and the supply curve cross: at that point both give the same Q and the same P. Take the ten miners from section 5, with inverse supply $P = 150 + 50Q$, and buyers with inverse demand $P = 750 - 50Q$.

Question 8.1: find the equilibrium

Curve 1 is $P = 750 - 50Q$. Curve 2 is $P = 150 + 50Q$. What are Q^* and P^*?

Step 1: which is which? Curve 1 falls: it is demand. Curve 2 rises: it is supply.

Step 2: set them equal. In equilibrium both give the same Q and the same P, so the two right-hand sides are equal:

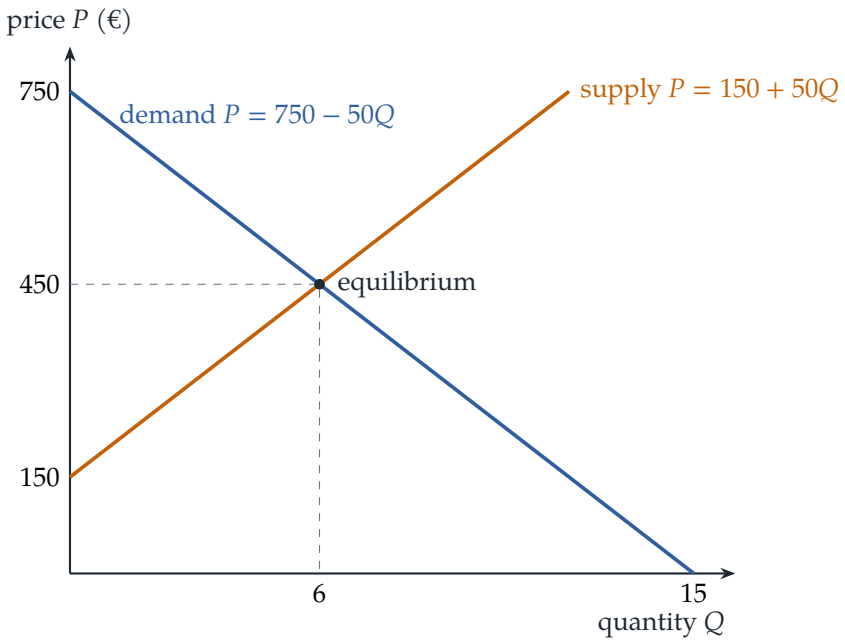
$$750 - 50Q = 150 + 50Q.$$

Step 3: solve for Q^*. $600 = 100Q$, so $Q^* = 6$.

Step 4: put Q^* into both curves. Demand: $750 - 50 \cdot 6 = 750 - 300 = 450$. Supply: $150 + 50 \cdot 6 = 150 + 300 = 450$. Both give the same P^*.

$Q^* = 6$, $P^* = €450$. Common mistakes: an arithmetic slip ($Q^* = 9$), using the supply intercept twice, or swapping P^* and Q^*.

At $Q^* = 6$ the marginal buyer is willing to pay €450 and the marginal miner (the 6th) needs exactly €450. The marginal buyer and the marginal seller pin down the price.



Demand and supply cross at $Q^* = 6$ and $P^* = €450$.

A general method. Write both curves in the same form, either both as $Q(P)$ or both as $P(Q)$. Set them equal and solve. Then put the solution into *both* curves: they must give the same answer, which is a check on your arithmetic. The deck gives four more questions.

Question 8.2: streaming subscriptions in Austria

P in € per month, Q in thousand subscribers. Demand is $Q_d = 300 - 10P$; supply is $Q_s = 5P - 60$. What are Q^* and P^*?

In equilibrium $Q_d = Q_s$: $300 - 10P = 5P - 60$. Collect P on one side: $360 = 15P$, so $P^* = 24$. (The 15 is the sum of the two slopes, $10 + 5$.) Then $Q^* = 300 - 10 \cdot 24 = 60$. Check with supply: $5 \cdot 24 - 60 = 60$.

$Q^* = 60$ (60,000 subscribers), $P^* = €24$.

Question 8.3: launches on reusable rockets

P in € million per launch, Q in launches per year. Demand is $P = 100 - 3Q$; supply is $P = 16 + Q$. What are Q^* and P^*?

Both are already in the form $P(Q)$: set them equal. $100 - 3Q = 16 + Q$, so $84 = 4Q$ and $Q^* = 21$. Put Q^* into both curves: $16 + 21 = 37$ and $100 - 3 \cdot 21 = 100 - 63 = 37$.

$Q^* = 21$ launches, $P^* = €37$ million.

Question 8.4: monthly memberships at Vienna climbing gyms

P in € per month, Q in hundred members. Demand is $Q_d = 192 - 4P$; inverse supply is $P = 12 + Q/2$. What are Q^* and P^*?

Same form first. $P = 12 + Q/2$ gives $Q/2 = P - 12$, so $Q_s = 2P - 24$. **Then $Q_d = Q_s$:** $192 - 4P = 2P - 24$, so $216 = 6P$ and $P^* = 36$. Then $Q^* = 192 - 4 \cdot 36 = 192 - 144 = 48$. Check: $2 \cdot 36 - 24 = 48$.

$Q^* = 48$ (4,800 members), $P^* = €36$.

Question 8.5: personal-trainer sessions at Vienna gyms

P in € per session, Q in hundred sessions per week. Demand is $Q_d = 70 - 2P$; inverse supply is $P = 15 + Q/2$. What is the equilibrium price?

Same form first. $P = 15 + Q/2$ gives $Q_s = 2P - 30$. **Then $Q_d = Q_s$:** $70 - 2P = 2P - 30$, so $100 = 4P$ and $P^* = 25$. Then $Q^* = 70 - 2 \cdot 25 = 20$. Check: $2 \cdot 25 - 30 = 20$.

$P^* = €25$ (and $Q^* = 20$, that is 2,000 sessions a week).

9 Shortage and surplus

What happens when the price is not the equilibrium price? We use the same market: demand $P = 750 - 50Q$, supply $P = 150 + 50Q$, and $P^* = €450$.

Shortage (excess demand)

Shortage (excess demand): at a price below P^*, buyers want more than sellers offer.

The story. The government caps the price at €350, below $P^* = €450$.

- **Buyers want:** $750 - 50Q = 350$, so $50Q = 400$ and $Q_d = 8$.
- **Miners mine:** $150 + 50Q = 350$, so $50Q = 200$ and $Q_s = 4$.
- **Shortage:** $Q_d - Q_s = 8 - 4 = 4$. Only 4 units are sold: the quantity traded is set by the short side of the market, here the sellers.

Without the cap, buyers would outbid each other and the price would rise.

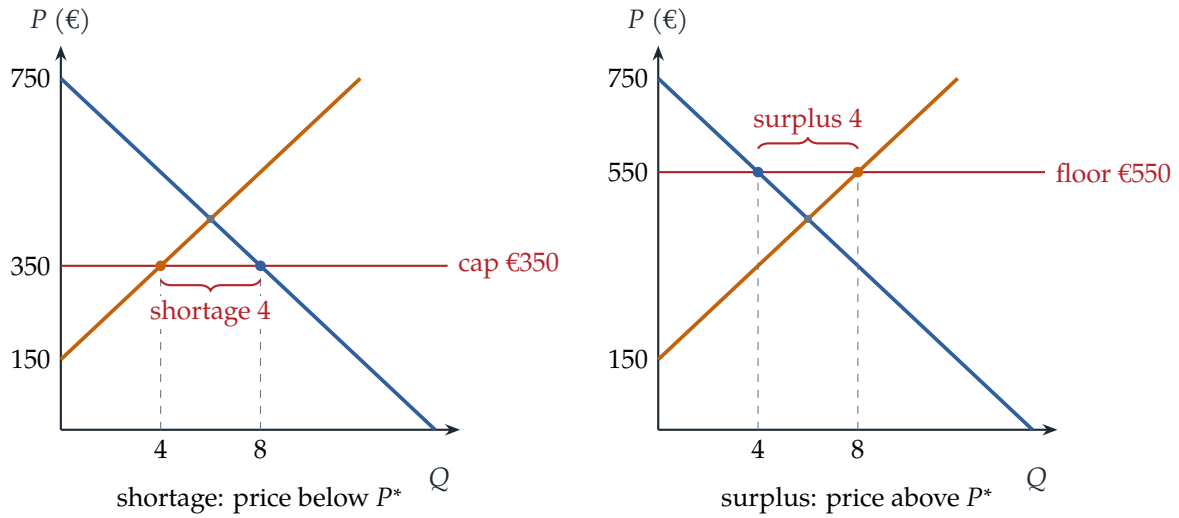
Surplus (excess supply)

Surplus (excess supply): at a price above P^*, sellers offer more than buyers want.

The story. The government sets a minimum price of €550, above $P^* = €450$.

- **Buyers want:** $750 - 50Q = 550$, so $50Q = 200$ and $Q_d = 4$.
- **Miners mine:** $150 + 50Q = 550$, so $50Q = 400$ and $Q_s = 8$.
- **Surplus:** $Q_s - Q_d = 8 - 4 = 4$. Only 4 units are sold: the short side is now the buyers.

Without the minimum price, sellers would cut their prices and the price would fall. (This "surplus" means excess supply. It is not the same as consumer surplus, which comes later in the course.)



Left: at the cap of €350, miners offer 4 (orange dot) and buyers want 8 (blue dot). Right: at the minimum price of €550, buyers want 4 and miners offer 8. In both cases only 4 units are sold. The grey dot is the equilibrium $(6, 450)$.

10 Summary

- **Demand / inverse demand.** Demand goes from a price to the quantity buyers want. Inverse demand goes from a quantity to the price buyers would pay: the marginal willingness to pay. Market demand is built by sorting buyers by reservation price; at a price, everyone whose reservation price is at least the price buys.
- **Supply / inverse supply.** Supply goes from a price to the quantity sellers offer. Inverse supply goes from a quantity to the price sellers need: the cost of the marginal seller.
- **Inverting and drawing.** Move the P-term to one side, then divide every term by its coefficient, keeping the sign. A demand line $Q = a - bP$ meets the axes at $P = a/b$ and $Q = a$; a supply line $Q = dP - c$ starts at $P = c/d$.
- **The price splits the gain.** The gain of a trade is the buyer's value minus the seller's cost. The total gain is the same at any price; the price decides who gets it.
- **One price for everyone.** In the synthetic market, buyers equal sellers at any price from €50 to €60. Then exactly the 5 best pairs trade, and the total gain is the maximum, €250.
- **Equilibrium.** Demand and supply give the same Q and the same P. The marginal buyer and the marginal seller pin down the price.
- **Shortage / surplus.** At a price below P^* buyers want more than sellers offer (shortage); above P^* sellers offer more than buyers want (surplus). In both cases only the quantity of the short side is sold.